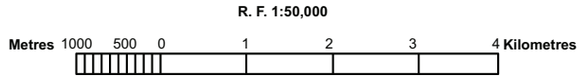
**Example:**




## Solved Examples

### Statement of Scale into R. F.

1. Convert the given Statement of Scale of 1 inch represents 5 miles into R. F.

   Solution

   The given Statement of Scale may be converted into R. F. using the following steps.

   1 inch represents 5 miles

   or 1 inch represents 5 × 63,360 inches

   (1 mile = 63,360 inches)

   or 1 inch represents 316,800 inches

   **Answer R. F. 1 : 316,800 or 1/316800**

### R. F. into Statement of Scale

2. Convert R. F of 1 : 200,000 into Statement of Scale (In Metric System)

   Solution

   The given R. F. of 1 : 200,000 may be converted into Statement of Scale using the following steps :

   1 : 200,000 means that 1 unit on the map represents 200,000 units on the ground.

   or 1 cm represents 200,000 /100,000 (1 km = 100,000 cm)

   or 1 cm represents 2 km

   **Answer 1 cm represents 2 km**

3. Convert the given Statement of Scale into Representative Fraction (R. F.).

   a. 5 cm represents 10 km

   b. 2 inches represents 4 miles

   c. 1 cm represents 100 metres

a. 5 cm represents 10 km

**Step 1:** convert into same units of measurement

(1 Km = 100000 cm)

**Step 2:** 10 km = 1000000 cm Therefore 5 cm: 1000000 cm

**Step 3:** simplify the ratio

1: 1000000/5

**Answer : R.F. = 1: 200000 or 1/200000**

b. 2 inches represents 4 miles

**Step 1:** convert into same units of measurement

(1 mile − 63,360 inches)

**Step 2:** 4 miles = 63,360 × 4 = 253440

Therefore 2 inches: 253440 inches

**Step 3:** simplify the ratio

1: 253440/2 = 126720

**Answer : R.F. = 1: 126720 or 1/126720**

c) 1 cm represents 100 metres

**Step 1:** convert into same units of measurement

(1m = 100 cm)

**Step 2:** 100 m = 10000 cm Therefore 1 cm: 10000 cm

**Answer : R.F. = 1: 10000 or 1/10000**

### Construction of the Graphical/Bar Scale

1. Construct a graphical scale for an R.F. 1 : 50,000 and read the distances in kilometre and metre.

(NOTE: By convention, a length of nearly 15 cm is taken to draw a graphical scale.)

**Calculations**

To get the length of line for the graphical scale, these steps may be followed:

R.F. =1 : 50,000 means that 1 unit of the map represents 50,000 units on the ground



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